Available online at www.sciencedirect.com

ScienceDirect

Current Opinion in Biotechnology

Next-generation genome-scale models for metabolic engineering

Zachary A King[1], Colton J Lloyd[1], Adam M Feist[1,2] and Bernhard O Palsson[1,2,3]

Constraint-based reconstruction and analysis (COBRA) methods have become widely used tools for metabolic engineering in both academic and industrial laboratories. By employing a genome-scale *in silico* representation of the metabolic network of a host organism, COBRA methods can be used to predict optimal genetic modifications that improve the rate and yield of chemical production. A new generation of COBRA models and methods is now being developed — encompassing many biological processes and simulation strategies — and next-generation models enable new types of predictions. Here, three key examples of applying COBRA methods to strain optimization are presented and discussed. Then, an outlook is provided on the next generation of COBRA models and the new types of predictions they will enable for systems metabolic engineering.

Addresses
[1] Department of Bioengineering, University of California, San Diego, 9500 Gilman Drive, La Jolla, CA 92093, USA
[2] Novo Nordisk Foundation Center for Biosustainability, Technical University of Denmark, 2800 Lyngby, Denmark
[3] Department of Pediatrics, University of California, San Diego, 9500 Gilman Drive, La Jolla, CA 92093, USA

Corresponding author: Palsson, Bernhard O (palsson@ucsd.edu)

Current Opinion in Biotechnology 2015, **35**:23–29

This review comes from a themed issue on **Chemical biotechnology**

Edited by **Uwe T Bornscheuer** and **Alex Toftgaard Nielsen**

http://dx.doi.org/10.1016/j.copbio.2014.12.016

0958-1669/© 2014 Elsevier Ltd. All rights reserved.

Introduction

The demand for raw material inputs to agriculture, industry, and energy are growing steadily, and concerns about environmental sustainability are becoming more acute; thus, alternatives to traditional, fossil-fuel based chemical production are becoming economically viable [1]. Cell factories, which use microorganisms to produce materials from renewable biomass, are an attractive alternative, and an increasing number of platform chemicals are being produced at industrial scale using engineered microorganisms [2]. To meet the demand for robust, high-yield production strains, an initial metabolic engineering strategy was developed, which used random mutagenesis and screening to identify strains with improved production performance (i.e., titer, productivity, and yield). However, the permutations possible in genomic sequences are so numerous that a mutagenesis and screening approach can only explore a small subset of possible strains, so the highest performance strains might never be identified. On the other hand, if *targeted* strain improvements can be predicted, then strains can be engineered which would not be found using untargeted mutagenesis and screening, and this can be accomplished with the tools of systems biology.

The goal of systems biology is to encode detailed information about an organism into a computational framework, with the goal of predicting cellular behavior from cellular genotype [3]. Thus, systems biology tools can provide targeted predictions of production strain modifications that improve their performance. The first major advancement toward this goal came with the development of the first genome-scale stoichiometric models of cellular metabolism. From these models emerged constraint-based reconstruction and analysis (COBRA) methods, with flux balance analysis (FBA) as a staple computational tool [4]. These advancements in turn led to the first studies in the field of systems metabolic engineering, where production strains are systematically built and improved through a combination of systems biology, synthetic biology, and evolutionary engineering [5–8].

An important first step in designing a production strain for a non-native metabolite is to identify and build a synthetic pathway [9]. COBRA methods have been employed successfully for pathway prediction and optimization [9,10]. After a pathway has been designed, strain optimization is performed to increase the yield and productivity of the strain. This commentary presents the most successful types of predictions made by COBRA tools for that process — optimizing microbial production strains — and then introduces a new generation of COBRA methods and prediction types which will advance systems metabolic engineering in the coming years.

Types of COBRA predictions used in strain optimization

A great number of COBRA methods have been developed [4], and the methods that have led to experimental

improvements in production strains can be categorized according to the types of predictions made. Three types of predictions have generally been employed: firstly, predicting maximum theoretical yield for native and non-native pathways, secondly, predicting gene deletions and their effects on a cell's ability to generate biomass and the target product, and thirdly, predicting the up and down expression of native genes necessary to improve product yield. Recent examples of strain engineering in each category are highlighted here.

Theoretical yield

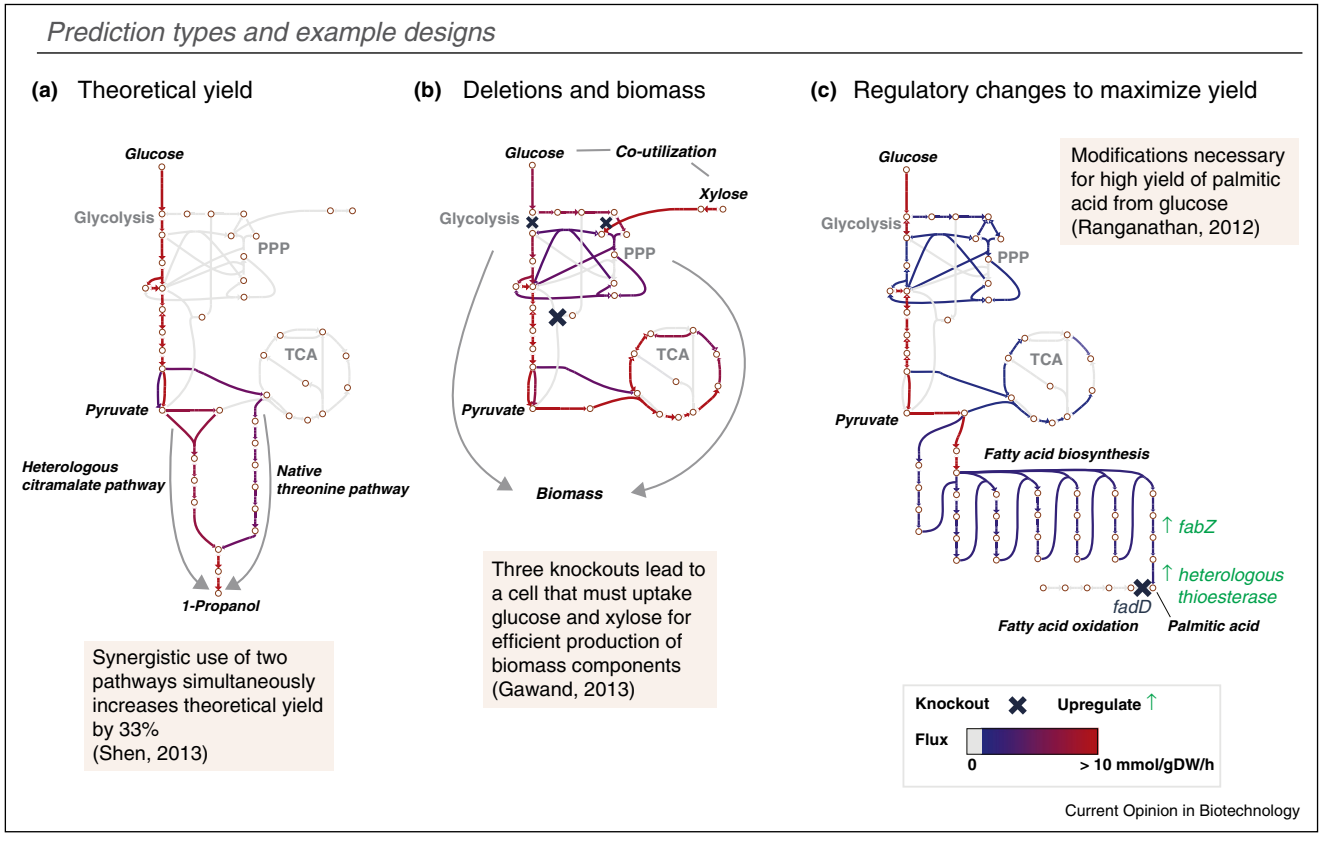
The most common prediction of COBRA methods in systems metabolic engineering has been the calculation of maximum *theoretical yield*, the percentage of substrate carbon that can be converted to a target molecule, given the limitations of carbon and redox balance in the stoichiometric network (Figure 1a). Yield is a critical consideration when determining the economic viability of chemical production, so these analyses have direct consequences for cell factory design.

Shen and Liao [11•] recently demonstrated the importance of theoretical yield calculations for optimizing production strains. In order to design a strain of *Escherichia coli* that produces 1-propanol, the authors used a simple mass-balanced and redox-balanced stoichiometric model and FBA to compare the theoretical yields of three routes to 1-propanol: the native threonine pathway, the non-native citramalate pathway, and a synergistic employment of both pathways (Figure 1a). The calculations revealed that the two pathways together have a theoretical yield of 1.33 mol 1-propanol per mol glucose, 33% higher than either individual pathway. Indeed, the authors constructed production strains for all three 1-propanol routes, and the synergistic employment of both pathways had the highest observed yield, ~30% greater yield than the citramalate pathway and ~55% greater yield than the threonine pathway.

Gene deletions and biomass requirements

Another class of COBRA predictions uses the biomass objective function — a representation of all the metabolite

Figure 1



Three types of COBRA predictions have been successfully implemented for systems metabolic engineering. **(a)** Theoretical yield. As an example, theoretical yield was maximized through synergistic use of two production pathways [11]. **(b)** Gene deletions and biomass. As an example, the SIMUP algorithm identifies three gene knockouts that force co-utilization of glucose and xylose to achieve maximum growth [19•]. **(c)** Regulatory changes to increase yield. As an example, gene up-regulations and deletions were used to increase the production of fatty acids [23•]. Figures were generated using Escher [49].

demands required for cell growth [12] — to predict how gene deletions will affect cellular phenotypes. For instance, a number of COBRA algorithms have been developed to identify groups of gene knockouts that are predicted to change the fermentation profile of a cell when growing at a maximum growth rate, a characteristic known as *growth-coupling* (e.g., OptKnock [13], OptGene [14], RobustKnock [15], OptSwap [16], for further discussion see [4]). However, only a few studies (e.g., [17,18]) have tested the validity of the predictions of these algorithms, and thus their significance to systems metabolic engineering is largely untested.

A similar COBRA method, called SIMUP, was recently shown to have direct usefulness for systems metabolic engineering [19•]. The SIMUP method predicted groups of gene knockouts that forced co-utilization of two substrates. Lignocellosic biomass is typically hydrolyzed into a mixture of glucose and xylose, but industrial organisms preferentially consume glucose over xylose. Thus, the SIMUP algorithm was used to identify a group of three gene knockouts that disable upper glycolysis and part of the pentose phosphate pathway so that both glucose and xylose consumption are necessary for rapid growth (Figure 1b). The results of the simulations were tested *in vitro*, and it was found that SIMUP accurately predicted strain designs that co-utilized these substrates, albeit with substrate uptake rates lower than the wild type.

Regulatory changes to increase yield

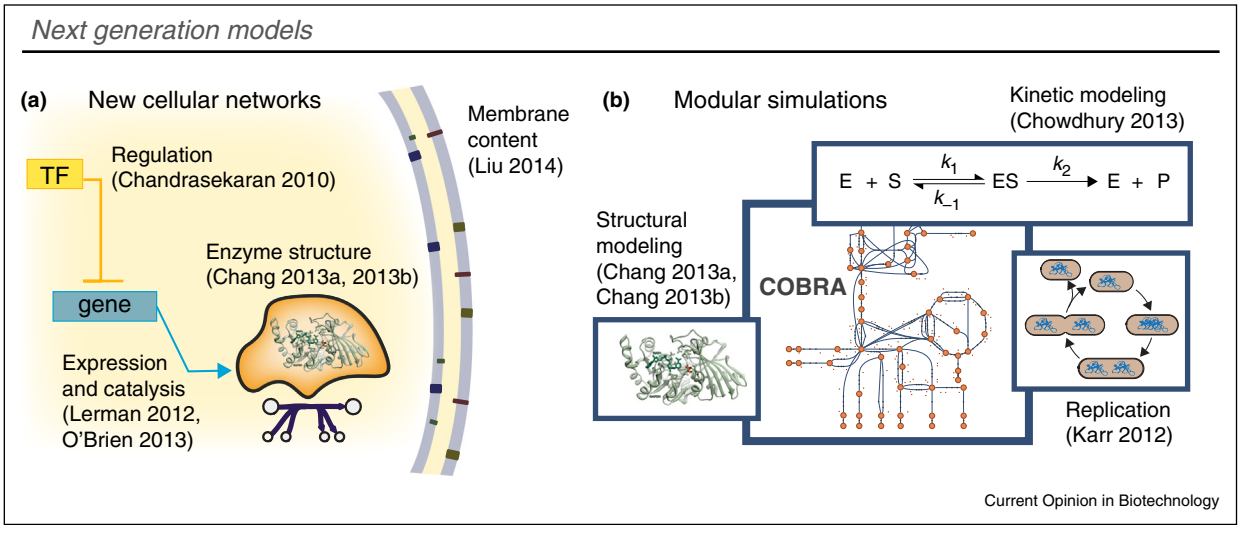
Some of the most successful COBRA methods for predicting modifications for systems metabolic engineering use empirical data (e.g., omics data) to generate a reference state for a host strain, then find the set of up-regulations, down-regulations, and/or knockouts necessary to increase the yield of a target molecule (e.g., MOMA [20], OptForce [21], FSEOF [22]). In a recent example of this approach, overproduction of C14–C16 fatty acids was engineered in *E. coli* by implementing the predicted gene up-regulations and deletions [23•] (Figure 1c). Similar strategies have been employed to produce polylactic acid (up to 11% yield by weight from glucose, using metabolic flux analysis and MOMA) [24], malonyl-CoA (fourfold increase in intracellular malonyl-CoA, using OptForce) [25], lycopene (greater than eightfold increase in lycopene concentration, using FSEOF and MOMA) [22], the antibiotic actinorhodin (52-fold increase in yield, using FSEOF) [26], and the recombinant protein human superoxide dismutase (1.4-fold increase in yield, using FSEOF and MOMA) [27•].

Thus, COBRA methods have been used to great effect in systems metabolic engineering, but the prediction types can be expanded. The tide is shifting toward a much greater range of predictive capabilities through use of more complex models that add additional constraints to the systematic analysis.

Next-generation models and predictions

As the applications of COBRA methods have multiplied, there has also been a continuous expansion in the scope and complexity of new models [28] and methods [4]. Together, the new methods deliver a vision of a COBRA modeling framework encompassing many biological networks and simulation strategies (Figure 2), so that many new types of predictions can be made (Table 1).

Figure 2



COBRA tools have advanced through **(a)** increased scope of cellular systems which can be modeled, and **(b)** highly modular simulation strategies, built upon genome-scale models of metabolism.

Table 1

Current and next-generation COBRA models — types of predictions that are possible.

Model scope	Model name	Number of components (reactions/metabolites)	Enabled predictions	References
Metabolism (M-Model)	*i*JO1366 (*E. coli*)	2583/1805	- Genetic manipulations for metabolite overproduction/consumption - Gene essentiality - Growth rate given constraining uptake rates - Metabolite secretion rates - Metabolic fluxes, considering stoichiometry of network, at given growth rate	[50]
Metabolism and gene expression (ME-Model)	*T. maritima* *i*OL1650-ME (*E. coli*)	17 535/18 209 76 414/56 902	- Maximum feasible growth rate without explicit constraining substrate uptake rates - Cellular macromolecular composition - Gene expression levels - Metabolic fluxes, considering enzyme cost, at maximum feasible growth rate - Nutrient limited phenotypes	[29] [32••]
ME-Model with cell membrane protein translocation	*i*JL1678-ME (*E. coli*)	79 871/70 751	- Membrane protein composition - Protein compartmentalization - Protein excretion rates	[32••]
Protein structural properties and metabolism	*T. Maritima* *E. coli* GEM-PRO	645/503 +478 enzyme structures iJO1366 +1268 enzyme structures	- Effect of enzyme structural characteristics on cell metabolism - Enzymes with low thermal stability which have greatest effect on cellular thermotolerance - Drug binding sites of enzymes and resultant cellular phenotype	[33] [34••,47]
Probabilistic transcriptional regulation and metabolism	*i*AF1260 PROM	M-Model +1773 regulatory interactions	- Quantitative impact on growth rate following genetic perturbation of transcriptional regulators - Downstream effects of gene up- and down-regulation on pathways effected by regulatory network	[38]
'Whole-cell' model	'Whole-cell' *M. genitalium*	Integration of 28 cellular process submodules, 525 genes	- Role of metabolism in cell-cycle regulation - Interaction of many cellular processes	[48•]

New cellular networks

Many of the latest improvements in COBRA modeling are based on collecting and reconstructing knowledge of cellular networks beyond metabolism (Figure 2a). These next-generation networks include gene product expression coupled to metabolism, protein translocation in the cell membrane, protein structures of metabolic enzymes, and transcriptional regulation. Each of these network reconstructions has been integrated with the metabolic network so that additional costs and constraints can be directly incorporated with existing COBRA methods.

The first model to integrate metabolism with gene expression (ME-model) was developed for the minimal thermophilic bacterium *Thermotoga maritima* [29], followed closely by the development of a ME-model for *E. coli* [30,31••]. With the inclusion of gene expression, ME-models have a host of new predictive capabilities. They directly account for the protein investment necessary for operating a metabolic pathway. In a metabolic model (M-model), enzymes are 'free;' a pathway of ten reactions has the same metabolic cost as a pathway of three reactions, as long as the overall stoichiometries of the pathways are equivalent. This often comes into play when simulating knockout mutant phenotypes that shift flux in exotic ways. In reality, longer pathways have a significantly greater enzyme production cost, and this cost is directly predicted in ME-model simulations. As an example of new predictions possible for systems engineering, the ME-model predicted acetate overflow metabolism as a result of rate-yield tradeoffs between metabolic pathways [31••]. Furthermore, the ME-model was shown to simulate batch growth conditions where the availability of enzyme protein limits the maximum substrate uptake rate and, therefore, the maximum growth rate. Both of these examples highlight important concepts in metabolic engineering that are captured by ME-models and not in models of metabolism.

The *E. coli* ME-model has been extended to include protein translocation in the cell membrane [32••]. In addition to the predictions described above, this extended ME-model can predict how spatial limitations in the inner and outer membranes (i.e., membrane crowding)

lead to tradeoffs between energy-efficient pathways that require membrane space (e.g., electron transport chain) and less-efficient pathways that require less membrane space (e.g., fermentation). These tradeoffs are not present in other COBRA models, and they could have major effects on metabolic engineering strategies that employ the expression of a large numbers of membrane-bound enzymes for substrate and product transport.

Another recent model expansion involved the collection of all the available enzyme structures for enzymes in the metabolic network of *T. maritima* [33], followed by the collection of enzyme structures for *E. coli* in a model called GEM-PRO [34••]. In addition to the application to thermotolerance discussed in the GEM-PRO publication, GEM-PRO can eventually be applied to exploring the multitude of effects that protein structures have on metabolic activity and regulation, including enzyme promiscuity, catalytic rates, complex formation, substrate channeling, and allosteric regulation [33,35,36].

A final area of active expansion in the scope of COBRA models is transcriptional regulation. This topic has received much attention [37], and so, in this article, it will only be noted that many strain engineering strategies are based on regulatory effects. Thus, techniques like the probabilistic regulation of metabolism (PROM) approach [38], which can predict the impact of transcriptional regulation on metabolic activity, will eventually allow for the direct prediction of regulatory modifications for strain engineering. A major limitation of PROM is that it requires large-scale empirical datasets to make accurate predictions. It is not yet possible to make forward predictions of transcriptional regulation using only the structure of the regulatory network. However, this kind of prediction will eventually be possible as the quality of binding information for transcription factors increases and the knowledgebase of regulatory interactions becomes more comprehensive [39–45].

Modularity

A second theme in the evolution of COBRA models is modularity of simulation strategies (Figure 2b). There have been long-standing challenges associated with genome-scale modeling of cellular networks using fine-grained approaches like stochastic and deterministic kinetic modeling, and so COBRA methods, which are tractable at the genome-scale, have risen in popularity [3]. By embracing modular simulations embedded in genome-scale COBRA models, it is possible to explore complex topics — including dynamics, concentrations, and physical structures — without losing genome-scale context.

While kinetic modeling is very difficult at the genome scale, the k-OptForce algorithm combines a genome-scale model of metabolism with a smaller-scale, deterministic kinetic model to identify strategies for metabolic engineering that incorporate knowledge about metabolite concentrations and kinetics [46••]. Additionally, the GEM-PRO model of metabolism and enzyme structure brings algorithms for protein structure analysis and simulation into the context of a COBRA model [34••,47].

Finally, a 'whole-cell' model of *Mycoplasma genitalium* has been reported, and this model includes modular simulations of all annotated gene functions in that minimal organism [48•]. The authors employ a highly modular platform where many types of simulations, including a COBRA model of metabolism, are executed at discrete time points. This model represents an extreme approach to modular simulation where every system can exist somewhat separately but with a shared notion of time. To manage this modularity computationally, all the constituent cellular processes are decoupled and simulated over a single time step (1 s), and then the modules are synchronized before the next time step is taken. A challenge associated with this strategy is that it is difficult to establish clear numerical convergence criteria.

Conclusion

With next-generation COBRA models, new predictions will be possible, ranging from enzyme promiscuity to allosteric regulation, from membrane crowding to overflow metabolism, and from metabolite concentration to cell cycle effects. To automate the design and creation of engineered cells, it will be necessary to incorporate detailed knowledge of interconnected cellular processes and to simulate these processes across time and length scales, using modular simulations. Furthermore, these new methods and models will need to be rigorously validated by repeatedly comparing predictions to experimental outcomes, in order to determine key modeling parameters and to close the gaps in our knowledge of biological systems. COBRA methods have proven their usefulness in a growing number of studies, and, as they expand to include many new cellular networks and types of simulation and prediction, the value and adoption of these methods are likely to grow.

Acknowledgments

Funding for this work was provided by the National Science Foundation Graduate Research Fellowship under Grant no. DGE-1144086 and by the Novo Nordisk Foundation.

References and recommended reading

Papers of particular interest, published within the period of review, have been highlighted as:

- • of special interest
- •• of outstanding interest

1. Johnson FX: *Industrial Biotechnology and Biomass Utilisation: Prospects and Challenges for the Developing World*. Vienna, Austria: United Nations Industrial Development Organization; 2007.

2. Manzer LE, van der Waal JC, Imhof P: **The industrial playing field for the conversion of biomass to renewable fuels and chemicals**. In *Catalytic Process Development for Renewable Materials.* Edited by van der Waal JC, Imhof P. Weinheim, Germany: Wiley-VCH; 2013:1-24.

3. Bordbar A, Monk JM, King ZA, Palsson BØ: **Constraint-based models predict metabolic and associated cellular functions**. *Nat Rev Genet* 2014, **15**:107-120.

4. Lewis NE, Nagarajan H, Palsson BØ: **Constraining the metabolic genotype–phenotype relationship using a phylogeny of in silico methods**. *Nat Rev Microbiol* 2012, **10**:291-305.

5. Kondo A, Ishii J, Hara KY, Hasunuma T, Matsuda F: **Development of microbial cell factories for bio-refinery through synthetic bioengineering**. *J Biotechnol* 2013, **163**:204-216.

6. Lee JW, Kim TY, Jang YS, Choi S, Lee SY: **Systems metabolic engineering for chemicals and materials**. *Trends Biotechnol* 2011, **29**:370-378.

7. Jang Y-S, Kim B, Shin JH, Choi YJ, Choi S, Song CW, Lee J, Park HG, Lee SY: **Bio-based production of C2–C6 platform chemicals**. *Biotechnol Bioeng* 2012, **109**:2437-2459.

8. Xu C, Liu L, Zhang Z, Jin D, Qiu J, Chen M: **Genome-scale metabolic model in guiding metabolic engineering of microbial improvement**. *Appl Microbiol Biotechnol* 2013, **97**: 519-539.

9. Shin JH, Kim HU, Kim DI, Lee SY: **Production of bulk chemicals via novel metabolic pathways in microorganisms**. *Biotechnol Adv* 2013, **31**:925-935.

10. Campodonico MA, Andrews BA, Asenjo JA, Palsson BO, Feist AM: **Generation of an atlas for commodity chemical production in *Escherichia coli* and a novel pathway prediction algorithm, GEM-Path**. *Metab Eng* 2014, **25**:140-158.

11. • Shen CR, Liao JC: **Synergy as design principle for metabolic engineering of 1-propanol production in *Escherichia coli***. *Metab Eng* 2013, **17**:12-22.

It was shown in *E. coli* that 1-propanol can be produced at a significantly higher rate (0.12 g/L/h) and yield (0.15 g/g of glucose) using both the native threonine pathway and a heterologous citramalate pathway than when using either of these to pathways individually (citramalate 0.11 g/g, 0.04 g/L/h; threonine 0.09 g/g, 0.04 g/L/h). Using FBA with a simple stoichiometric network, the authors were able to predict that synergy between these two pathways allowed for the observed increase in 1-propanol production.

12. Feist AM, Zielinski DC, Orth JD, Schellenberger J, Herrgard MJ, Palsson BØ: **Model-driven evaluation of the production potential for growth-coupled products of *Escherichia coli***. *Metab Eng* 2010, **12**:173-186.

13. Burgard AP, Pharkya P, Maranas CD: **Optknock: a bilevel programming framework for identifying gene knockout strategies for microbial strain optimization**. *Biotechnol Bioeng* 2003, **84**:647-657.

14. Patil KR, Rocha I, Förster J, Nielsen J: **Evolutionary programming as a platform for in silico metabolic engineering**. *BMC Bioinform* 2005, **6**:308.

15. Tepper N, Shlomi T: **Predicting metabolic engineering knockout strategies for chemical production: accounting for competing pathways**. *Bioinformatics* 2010, **26**:536-543.

16. King ZA, Feist AM: **Optimizing cofactor specificity of oxidoreductase enzymes for the generation of microbial production strains—OptSwap**. *Ind Biotechnol* 2013, **9**:236-246.

17. Fong SS, Burgard AP, Herring CD, Knight EM, Blattner FR, Maranas CD, Palsson BO: **In silico design and adaptive evolution of *Escherichia coli* for production of lactic acid**. *Biotechnol Bioeng* 2005, **91**:643-648.

18. Yim H, Haselbeck R, Niu W, Pujol-Baxley C, Burgard A, Boldt J, Khandurina J, Trawick JD, Osterhout RE, Stephen R *et al.*: **Metabolic engineering of *Escherichia coli* for direct production of 1,4-butanediol**. *Nat Chem Biol* 2011, **7**: 445-452.

19. • Gawand P, Hyland P, Ekins A, Martin VJJ, Mahadevan R: **Novel approach to engineer strains for simultaneous sugar utilization**. *Metab Eng* 2013, **20**:63-72.

In order to design a more efficient industrial *E. coli* strain, the SIMUP algorithm was developed to suggest gene knockouts that would force *E. coli* to grow using both xylose and glucose simultaneously. The designed mutants were shown to successfully display this growth phenotype *in vivo*.

20. Segre D, Vitkup D, Church GM: **Analysis of optimality in natural and perturbed metabolic networks**. *Proc Natl Acad Sci* 2002, **99**:15112-15117.

21. Ranganathan S, Suthers PF, Maranas CD: **OptForce: an optimization procedure for identifying all genetic manipulations leading to targeted overproductions**. *PLoS Comput Biol* 2010, **6**:e1000744.

22. Choi HS, Lee SY, Kim TY, Woo HM: **In silico identification of gene amplification targets for improvement of lycopene production**. *Appl Environ Microbiol* 2010, **76**:3097-3105.

23. • Ranganathan S, Wei Tee T, Chowdhury A, Zomorrodi AR, Moon Yoon J, Fu Y, Shanks JV, Maranas CD: **An integrated computational and experimental study for overproducing fatty acids in *Escherichia coli***. *Metab Eng* 2012, **14**:687-704.

Experimental data was incorporated into a genome scale *E. coli* metabolic model in order to calculate a wild-type flux state for the metabolic network. Using the OptForce algorithm, which determines reactions within the metabolic network that must be increased or decreased in order to meet a guaranteed minimum flux toward the target metabolite, a strain was engineered that produces C14–C16 fatty acids at ~39% of the maximum theoretical yield.

24. Jung YK, Kim TY, Park SJ, Lee SY: **Metabolic engineering of *Escherichia coli* for the production of polylactic acid and its copolymers**. *Biotechnol Bioeng* 2010, **105**:161-171.

25. Xu P, Ranganathan S, Fowler ZL, Maranas CD, Koffas MAG: **Genome-scale metabolic network modeling results in minimal interventions that cooperatively force carbon flux towards malonyl-CoA**. *Metab Eng* 2011, **13**:578-587.

26. Kim M, Sang Yi J, Kim J, Kim J-N, Kim MW, Kim B-G: **Reconstruction of a high-quality metabolic model enables the identification of gene overexpression targets for enhanced antibiotic production in *Streptomyces coelicolor* A3(2)**. *Biotechnol J* 2014, **9**:1185-1194.

27. • Nocon J, Steiger MG, Pfeffer M, Sohn SB, Kim TY, Maurer M, Rußmayer H, Pflügl S, Ask M, Haberhauer-Troyer C *et al.*: **Model based engineering of *Pichia pastoris* central metabolism enhances recombinant protein production**. *Metab Eng* 2014, **24**:129-138.

Heterologous pathways for cytosolic human superoxide dismutase and β-glucuronidase production were inserted into the yeast *Pichia pastoris* metabolic model. Two methods were used for cell factory optimization: FSEOF (Flux Scanning based on Enforced Objective Function) simulates an increase in production of the target metabolite and determines which fluxes within the metabolic network must change as a result, and MOMA (Minimization of Metabolic Adjustment) predicts the metabolic state of the organism following a genetic perturbation by minimizing the deviation from the wild-type state. These methods were used to predict gene targets for up-regulation and deletion, and implementing the predictions resulted in multiple strains with increased production of human superoxide dismutase and β-glucuronidase.

28. Feist AM, Herrgård MJ, Thiele I, Reed JL, Palsson BØ: **Reconstruction of biochemical networks in microorganisms**. *Nat Rev Microbiol* 2009, **7**:129-143.

29. Lerman JA, Hyduke DR, Latif H, Portnoy VA, Lewis NE, Orth JD, Schrimpe-Rutledge AC, Smith RD, Adkins JN, Zengler K *et al.*: **In silico method for modelling metabolism and gene product expression at genome scale**. *Nat Commun* 2012, **3**:929.

30. Thiele I, Fleming RMT, Que R, Bordbar A, Diep D, Palsson BO: **Multiscale modeling of metabolism and macromolecular synthesis in *E. coli* and its application to the evolution of codon usage**. *PLoS ONE* 2012, **7**:e45635.

31. •• O'Brien EJ, Lerman JA, Chang RL, Hyduke DR, Palsson BØ: **Genome-scale models of metabolism and gene expression extend and refine growth phenotype prediction**. *Mol Syst Biol* 2013, **9**:693.

A predictive model integrating enzyme expression machinery with the metabolic network reconstruction (ME-model) of *E. coli* was developed, which computes ~80% of the mass of the cell proteome. The predictive capabilities of the new model in comparison to M-models were demonstrated. Cell composition was shown to vary as a function of growth rate under the constraints of the ME-model, and varying phenotypes between substrate limited and substrate excess conditions were explored, leading to identification of key regulatory modules.

32. •• Liu J, O'Brien E, Lerman J, Zengler K, Palsson BØ, Feist AM: **Reconstruction and modeling protein translocation and compartmentalization in *Escherichia coli* at the genome-scale**. *BMC Syst Biol* 2014, **8**.

The ME-model of *E. coli* was extended to include protein translocation and cell membrane composition. This model was able to predict proteome content with high accuracy, compute tradeoffs between spatial requirements in the cell membrane and pathway efficiency for metabolic pathways, and show the effects of membrane enzyme efficiencies on cell growth.

33. Zhang Y, Thiele I, Weekes D, Li Z, Jaroszewski L, Ginalski K, Deacon AM, Wooley J, Lesley SA, Wilson IA *et al.*: **Three-dimensional structural view of the central metabolic network of *Thermotoga maritima***. *Science* 2009, **325**:1544-1549.

34. •• Chang RL, Andrews K, Kim D, Li Z, Godzik A, Palsson BO: **Structural systems biology evaluation of metabolic thermotolerance in *Escherichia coli***. *Science* 2013, **340**: 1220-1223.

Structural information about catalytic enzymes was used to parameterize a genome-scale model of *E. coli*. By incorporating structural properties into the metabolic model, it was possible to determine which enzymes within the metabolic network had the greatest influence on the thermostability of the organism. Furthermore, it was shown that the thermotolerance of *E. coli* could be increased by supplementing the growth media with metabolites directly downstream of thermosensitive enzymes.

35. Beltrao P, Kiel C, Serrano L: **Structures in systems biology**. *Curr Opin Struct Biol* 2007, **17**:378-384.

36. Fisher AK, Freedman BG, Bevan DR, Senger RS: **A review of metabolic and enzymatic engineering strategies for designing and optimizing performance of microbial cell factories**. *Comput Struct Biotechnol J* 2014, **11**:91-99.

37. Gonçalves E, Bucher J, Ryll A, Niklas J, Mauch K, Klamt S, Rocha M, Saez-Rodriguez J: **Bridging the layers: towards integration of signal transduction, regulation and metabolism into mathematical models**. *Mol Biosyst* 2013, **9**:1576-1583.

38. Chandrasekaran S, Price ND: **Probabilistic integrative modeling of genome-scale metabolic and regulatory networks in *Escherichia coli* and *Mycobacterium tuberculosis***. *Proc Natl Acad Sci USA* 2010, **107**:17845-17850.

39. Cho B-K, Federowicz S, Park Y-S, Zengler K, Palsson BØ: **Deciphering the transcriptional regulatory logic of amino acid metabolism**. *Nat Chem Biol* 2012, **8**:65-71.

40. Seo SW, Kim D, Latif H, O'Brien EJ, Szubin R, Palsson BO: **Deciphering Fur transcriptional regulatory network highlights its complex role beyond iron metabolism in *Escherichia coli***. *Nat Commun* 2014, **5**:4910.

41. Cho B-K, Kim D, Knight EM, Zengler K, Palsson BO: **Genome-scale reconstruction of the sigma factor network in *Escherichia coli*: topology and functional states**. *BMC Biol* 2014, **12**:4.

42. Carrera J, Estrela R, Luo J: **An integrative, multi-scale, genome-wide model reveals the phenotypic landscape of *Escherichia coli***. *Mol Syst Biol* 2014, **10**:1-13.

43. Lee SG, Park JH, Hou BK, Kim YH, Kim CM, Hwang KS: **Effect of weight-added regulatory networks on constraint-based metabolic models of *Escherichia coli***. *Biosystems* 2007, **90**: 843-855.

44. Cho B-K, Charusanti P, Herrgård MJ, Palsson BO: **Microbial regulatory and metabolic networks**. *Curr Opin Biotechnol* 2007, **18**:360-364.

45. Federowicz S, Kim D, Ebrahim A, Lerman J, Nagarajan H, Cho B, Zengler K, Palsson B: **Determining the control circuitry of redox metabolism at the genome-scale**. *PLoS Genet* 2014, **10**:e1004264.

46. •• Chowdhury A, Zomorrodi AR: **Maranas CD: k-OptForce: integrating kinetics with flux balance analysis for strain design**. *PLoS Comput Biol* 2014, **10**:e1003487.

The k-OptForce algorithm combines a genome-scale model of metabolism with a smaller-scale, deterministic kinetic model to identify strategies for metabolic engineering that incorporate knowledge about metabolite concentrations and kinetics. This method was used to study L-threonine production in *E. coli* and triacetic acid lactone production in *S. cerevisiae*.

47. Chang RL, Xie L, Bourne PE, Palsson BO: **Antibacterial mechanisms identified through structural systems pharmacology**. *BMC Syst Biol* 2013, **7**:102.

48. • Karr JR, Sanghvi JC, Macklin DN, Gutschow MV, Jacobs JM, Bolival B, Assad-Garcia N, Glass JI, Covert MW: **A whole-cell computational model predicts phenotype from genotype**. *Cell* 2012, **150**:389-401.

A 'whole-cell' model of *Mycoplasma genitalium* was reported, representing 525 genes and 28 subsystems [47]. The authors employ a highly modular platform where each subsystem, including a COBRA model of metabolism, is executed at discrete time points, followed by reconciliation before the next time step. The model was used to explore protein–DNA association rates, the relationship between DNA replication and initiation, and to direct experimental elucidation of kinetic parameters.

49. King ZA, Ebrahim A: *Escher (Version 1.0) [Software]*. Available from https://escher.github.io 2014.

50. Orth JD, Conrad TM, Na J, Lerman JA, Nam H, Feist AM, Palsson BØ: **A comprehensive genome-scale reconstruction of *Escherichia coli* metabolism—2011**. *Mol Syst Biol* 2011, **7**:535.